Hardening the weather pipeline to Improve accuracy in crop yield forecasting

Preamble

The success of GaiaDhi's AgTech platform Holos is centered around increasing the accuracy of crop yield forecasting. Holos uses crop simulation models which base crop growth on crop science and inputs related to environment, and farming practices. The higher the accuracy of the inputs, the higher the accuracy of the models.

One of the highly dynamic, and hard to forecast environmental inputs is weather (key parameters: max temp, min temp, solar radiation, and rainfall), and they differ location to location, and season to season.

Project Details

Current Holos Weather Pipeline

Holos uses NASA Power API to obtain current and historical weather at any latitude and longitude. Historical weather data as part of crop yield forecast for the upcoming season. While NASA Power is very flexible, we have been informed it is not localized and may also vary seasonally with localized observation.

IMD Resolution	
Spatial - coarse	Tmin and Tmax at 1 deg (111 km x 111 km), Rain at 0.25 deg (28km x 28km), no SRAD
Temporal	1951 - 2020
NASA POWER Resolution	
Spatial - fine	any, incl SRAD
Temporal	1985 - current

In an attempt to localise, Holos does the following

1. IMD has provided low resolution observed (ground weather stations and interpolated). The assumption is that this data is ground truth. IMD does not provide solar radiation.
2. Holos has a precomputed database of deltas between NASA Power data and IMD (delta = IMD - NASA POWER), for points (lat, lon) where available. Call these "anchor points". The deltas are called "bias"es.

3. At time of simulation: for a point of simulation, which may vary from an anchor point, a bias correction is applied:
    1. NASA Power for that point is obtained
    2. Anchor point closest to simulation point is determined, and the delta for that point is extracted
    3. Weather for that point = NASA POWER weather for that point + delta. For a point where IMD data is available, this bias correction would end up at the IMD value
4. By bias correcting NASA Power to observed and interpolated IMD localised data, we expect to have localised NASA Power.

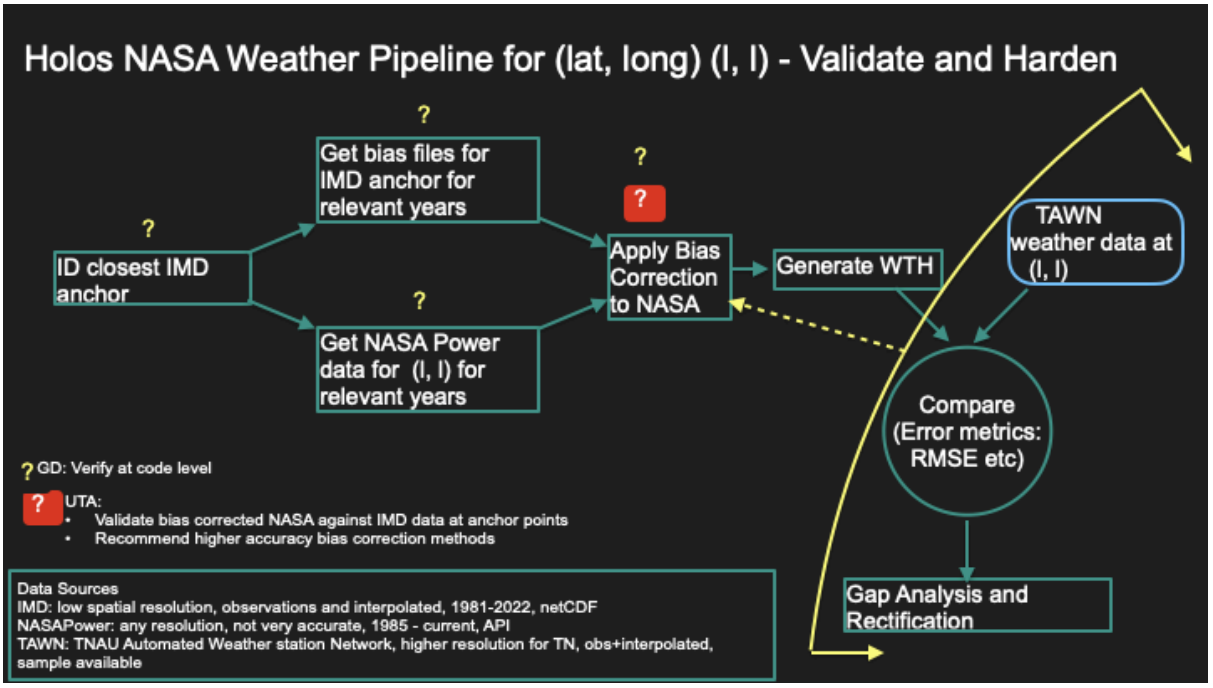
High Level Problem Statement

Goal is to (unless/until other higher accuracy sources are identified) use NASA Power as it is API driven and has data at any point. We need to find the correct data set (IMD/TAWN for now) to bias with and the bias function.

Priority 1: Current Implementation: Based on IMD

This can be acted upon immediately, as no further data is needed.

1. Is bias correction working as expected? We have not code verified nor validated the stages in the weather pipeline.
    a. GD: Code verification can be done
    b. UTA:
        i. Is subtraction as the current bias correction function, the right one to use?
        ii. At the point where IMD data is available, is bias corrected NASA POWER the same as IMD data? If it is not working, fix it OR not?
2. We also want to find out if there is a pattern where unlocalized NASA Power varies from IMD. This will resolve the “myth”. For instance:
    a. Are there regions where the gaps exist and/or are there specific seasons where the gaps exist?
    b. Which of the four weather parameters shows the most gap and in which area/season?



Priority 2: Availability of Higher Resolution TAWN data

TNAU has a set of automated weather stations in about 385 (http://tawn.tnau.ac.in/General/HomePublicUI.aspx) locations in Tamil Nadu, and interpolated to cover 3000+ poimts. GD is working with TNAU to obtain historical, current, and their WRF based localised predictions.

Why consider TAWN?

1. Ground truth with higher spatial resolution: Many more points (lat, long) covered than IMD
2. Contains more interpolated points to validate NASA Power.
3. Supposedly more accurate than IMD
4. Has SRAD
5. More importantly has soil moisture, which will be of use to assimilate into crop model (out of scope of this project).

Open Questions

Pending analysis of TAWN data to be provided:

1. Do we replace IMD with TAWN for TN?
2. Do we replace NASA Power for TN?
3. Is there an all India AWS network (http://aws.imd.gov.in:8091/) across states?

Proposal to use TAWN:

Replace IMD for TN with TAWN, build bias based on TAWN for bias correcting NASA Power at simulation time. While this is out of scope of this project, following should be:

1. Analyse gaps between NASA Power and TAWN data
2. Identify a “better” bias correction function

Priority 3 or 4: FarmAgain ClimateSense

Pending engagement w FarmAgain, we could try to obtain weather data from their instruments which are deployed in farms in India

Pros: Probably quick to obtain

Cons: Limited to farms in certain states - not too broad, across regions.

Priority 4 or 3: ML Models that are based on Physical Models

Google's GraphCast which is based on outputs from ECMWF's physical models.

Pros:

1. ECMWF is a well recognized weather model
2. GraphCast claims to be more accurate than ECMWF
3. If global, use it as a global bias source, rather than just IMD for India and another local source for another country?

Cons:

1. Same coarse resolution as IMD
2. Does this need to be localised? This will assess the global bias source applicability.

Scope for UTA:

1. Identify where GraphCast output can be downloaded
2. Compare against NASA Power across select regions
3. Compare against IMD and TAWN for India.
4. Check on availability of localised data across multiple regions
5. Build a bias function to be used with NASA Power

Davis Instruments

https://www.davisinstruments.com/pages/data-collection
Scope for UTA Project: None

Defining Project Boundaries

To keep focus during project execution:

1. Geo: Localization/regions. Narrow to one or two regions for the purpose of this project. At the moment, access to data of higher spatial resolution in the Tamil Nadu region is expected to be available.
2. Temporal: Seasonal. Start with one and expand to another
3. Prediction
    a. Physical models
    a. ML models based on physical models
        . GraphCast
4. Scope of project:
    a. Geos: Texas, Cauvery basin
    a. [is this of importance?] Crops: short term (< 180 days): rice, wheat, alfalfa, cotton
    a. Time: for next x days, with adjustable confidence

High Level Project Steps

To be discussed and decided. Will involve ML colleague ASAP.

Goal is to (unless/until other higher accuracy sources are identified) use NASA Power as it is API driven and has data at any point. We need to find the correct data set to bias with and the bias function.

Question: How do you improve (ie., localize) one data set (ie NASA Power) based on another data set (ie., IMD or TAWN or XYZ - which is the observed local data)?

1. Understand problem space and requirement
2. Spatial grid of IMD data set
3. For a constant spatial resolution, predict temporarily (Benefit: we could get rid of using IMD all together and reference year; caveat: only for anchor points - but could be repeated for another region if ground truth is available)
    a. Time-Series on a seasonal basis of IMD data set
    b. Time-Series on a seasonal basis of NASA power of the same (anchor) points in IMD (ie., keep spatial resolution a constant)
    c. Identify and analyse gaps between NASA and IMD
    d. Develop an ML model to temporarily predict the NASA set for anchor points only
4. TAWN
    a. Analyse time-series between spatially extended NASA Power ML model (above)
    b. Use TAWN, increase the accuracy of spatial interpolation in NASA
    c. Can TAWN be used in step 3d)?